

# The 3,371 semiaffine orbits of order-four anti-coordinate grid decompositions

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## Abstract

We classify pairs of orthogonal partitions of the sixteen points of  $AG(2, 4)$  into four four-point blocks, subject to the condition that every block fails to be a transversal of each parallel class. Under the full semiaffine point group and exchange of the two partitions, there are exactly 3,371 orbits. We supply a canonical representative and every stabilizer element for each orbit in a machine-readable supplement. The stabilizer orders are 1, 2, 4, 8, 16, occurring in 3, 229, 110, 23, 7, and 2 orbits, respectively. The proof combines exhaustive finite enumeration, complete group-action checks, and an independent count of all 18,954,720 admissible unordered pairs. The result concerns this precise finite classification; historical novelty is not asserted.

## 1 Objects, equivalence and the classification

Put  $\mathbb{F}_4 = \mathbb{F}_2[t]/(t^2 + t + 1)$  and  $V = \mathbb{F}_4^2$ . A direction is a one-dimensional  $\mathbb{F}_4$ -subspace of  $V$ ; its four cosets form a parallel class of affine lines. There are five directions. A four-point subset  $B$  is *eligible* if, for every parallel class,  $B$  is not a transversal of that class. Thus the condition is imposed separately for each of the five directions.

Let  $\mathcal{P}$  be the set of partitions of  $V$  into four eligible blocks. Blocks in a partition are unlabeled. Two partitions  $\mathcal{R}, \mathcal{C}$  are *compatible* when

$$|R \cap C| = 1 \quad (R \in \mathcal{R}, C \in \mathcal{C}). \quad (1)$$

These intersections give a four-by-four grid on  $V$ . Write  $\mathcal{S}$  for the set of unordered compatible pairs  $\{\mathcal{R}, \mathcal{C}\}$ . The group  $G = \text{AGL}(2, 4)$  acts simultaneously on all points. In this paper, an anti-coordinate grid decomposition means exactly an element of  $\mathcal{S}$ . Passing to unordered pairs implements the allowed exchange of the two partitions; block relabelings are already absent from the objects.

**Theorem 1.** *The action of  $G$  on  $\mathcal{S}$  has exactly 3,371 orbits, and  $|\mathcal{S}| = 18,954,720$ . The 3,371 entries of the supplementary file `evidence/pair-orbits.json` give one lexicographically canonical representative of each orbit and its complete stabilizer, with the encodings of Section 4. The orbit census is Table 1, and no other stabilizer orders occur.*

The motivating problem identifies Aoyama, Hasegawa, Hayashi and Sagara's *Mutually orthogonal anti-Latin squares* [1] as its closest literature lead. The classification requested here is the full compatible-partition census, including representatives and stabilizers. The cited primary text and its reported theorem attributions were not verified in this work, so we do not claim that this census was absent from it. No result from that reference is used in the proof. As adjacent computational methodology, proof-carrying graph reductions [2] also concern finite orbit enumeration and replayable certificates; that citation is not a geometric classification theorem used here.

| Stabilizer order | Number of orbits | Size of each orbit |
|------------------|------------------|--------------------|
| 1                | 3,229            | 5,760              |
| 2                | 110              | 2,880              |
| 4                | 23               | 1,440              |
| 8                | 7                | 720                |
| 16               | 2                | 360                |

Table 1: The complete census for unordered partition pairs.

## 2 Finite domain and group action

**Lemma 2.** *A four-point block is eligible if and only if its six unordered point pairs realize all five directions.*

*Proof.* Fix a direction  $D$ . The four lines of its parallel class partition  $V$ . Four points form a transversal precisely when they occupy four distinct lines. Failure of this property is equivalent to two of the points lying on the same line, that is, to their difference having direction  $D$ . Apply this equivalence to all five directions.  $\square$

This gives a short exhaustive construction of  $\mathcal{P}$ . Test all  $\binom{16}{4} = 1,820$  four-point subsets for eligibility. To form a partition, choose a block containing the least uncovered point, remove that block, and recurse until no point remains. Each partition has exactly one block through the least uncovered point at every stage, so this recursion generates every partition exactly once. The executed computation gives 720 eligible blocks and 66,960 partitions.

The verification uses a different construction. It builds the twenty affine lines explicitly and tests intersection sizes with each parallel class. It then groups disjoint pairs of eligible blocks by their eight-point union  $M$ . A pair with union  $M$  and a pair with union  $V \setminus M$  together form a partition. Considering one ordering of these two complementary unions, then sorting and removing duplicate four-block tuples, gives exactly  $\mathcal{P}$ : every partition occurs through the three pairings of its four blocks. This also gives 66,960 partitions.

The elements of  $G$  have the form

$$(x, y) \mapsto (ax^{2^e} + by^{2^e} + u, cx^{2^e} + dy^{2^e} + v), \quad (2)$$

where  $a, b, c, d, u, v \in \mathbb{F}_4$ ,  $ad - bc \neq 0$ , and  $e \in \{0, 1\}$ . There are 15 possible nonzero first matrix columns and 12 choices for the second column outside their span. Hence

$$|G| = 16(16 - 1)(16 - 4) \cdot 2 = 5,760. \quad (3)$$

The parameter tuples give distinct maps. The image of zero determines the translation; the two automorphisms are distinct because Frobenius is not  $\mathbb{F}_4$ -linear. For a fixed automorphism the images of a basis determine the matrix. Semiaffine maps permute the affine lines and parallel classes and preserve intersection cardinalities. They therefore act on  $\mathcal{P}$  and  $\mathcal{S}$ .

The independent verifier constructs the same group by closing the following permutations under composition:

$$\begin{aligned} &\text{translations by } (1, 0), (t, 0), (0, 1), (0, t), \\ &(x, y) \mapsto (tx, y), \quad (x, y) \mapsto (y, x), \quad (x, y) \mapsto (x + y, y), \quad (x, y) \mapsto (x^2, y^2). \end{aligned}$$

All generators belong to  $G$ . The computed closure has 5,760 distinct permutations, so by (3) it is all of  $G$ . The verifier also checks that every resulting map preserves the set of eligible blocks.

### 3 Counting the entire admissible domain

**Lemma 3.** *For a fixed  $\mathcal{R} \in \mathcal{P}$ , its compatible partners are enumerated without repetition by a subset of the  $(4!)^3$  triples of permutations of four points.*

*Proof.* Order the four blocks of  $\mathcal{R}$  and the four points in each block in any fixed way. Label the blocks of a compatible  $\mathcal{C}$  by their unique points in the first block of  $\mathcal{R}$ . In each of the other three blocks of  $\mathcal{R}$ , assigning its points to these four column labels is a permutation. Conversely, a triple of permutations forms four disjoint four-point column blocks covering  $V$ , each meeting every row block once. It gives a member of  $\mathcal{P}$  exactly when all four column blocks are eligible. These constructions are inverse.  $\square$

The verifier performs this test for every one of the 66,960 partitions. It tests exactly

$$66,960 \cdot 24^3 = 925,655,040$$

assignments and accepts 37,909,440. This count uses neither the proposed pair representatives nor their stabilizers. Compatibility is symmetric and excludes  $\mathcal{R} = \mathcal{C}$ , since a block intersects itself in four points. It follows that

$$|\mathcal{S}| = 37,909,440/2 = 18,954,720. \quad (4)$$

For a compact independently checkable subtotal, the 17 partition orbits and their partner counts are given in Table 2. If  $w_i$  is the orbit size and  $d_i$  the number of partners of its representative, invariance gives  $\sum_i w_i = 66,960$  and  $\sum_i w_i d_i = 37,909,440$ . The direct all-partition count above does not assume this reduction.

| Index | Partition orbit size $w_i$ | Partners $d_i$ | Ordered-pair orbits |
|-------|----------------------------|----------------|---------------------|
| 0     | 720                        | 464            | 73                  |
| 1     | 1,440                      | 556            | 157                 |
| 2     | 5,760                      | 503            | 503                 |
| 3     | 5,760                      | 541            | 541                 |
| 4     | 5,760                      | 513            | 513                 |
| 5     | 1,440                      | 463            | 120                 |
| 6     | 5,760                      | 588            | 588                 |
| 7     | 1,440                      | 619            | 160                 |
| 8     | 2,880                      | 670            | 335                 |
| 9     | 5,760                      | 558            | 558                 |
| 10    | 2,880                      | 478            | 255                 |
| 11    | 5,760                      | 581            | 581                 |
| 12    | 5,760                      | 505            | 505                 |
| 13    | 5,760                      | 676            | 676                 |
| 14    | 2,880                      | 698            | 365                 |
| 15    | 5,760                      | 594            | 594                 |
| 16    | 1,440                      | 528            | 138                 |
| Total | 66,960                     | —              | 6,662               |

Table 2: Partition-orbit indices match the supplementary `partition-orbits.json`. The final column counts ordered-pair  $G$ -orbits whose first partition lies in the indicated orbit.

A separately implemented check counts partners by complementary unions. For fixed  $\mathcal{R}$ , let  $T$  be the eligible blocks meeting every block of  $\mathcal{R}$  once. Let  $h(M)$  count unordered disjoint pairs of members of  $T$  with union  $M$ . Using integer masks to order sets,

$$d(\mathcal{R}) = \frac{1}{3} \sum_{M < V \setminus M} h(M)h(V \setminus M). \quad (5)$$

Each product constructs a compatible partition, and each such partition has exactly three pairings into two unordered pairs of blocks. This proves the divisor three. Evaluation on the 17 partition representatives, with independently checked orbit sizes, gives the same weighted total.

## 4 Representatives, stabilizers and completeness

Encode  $0, 1, t, t+1$  by  $0, 1, 2, 3$ . Encode  $(x, y)$  by  $x + 4y$ , where the arithmetic in this encoding uses the integer codes. A block is the integer mask whose bit at position  $p$  is one exactly when it contains point  $p$ . Encode a partition by its increasing four-tuple of masks, and an unordered pair by its increasing pair of partition tuples. The canonical representative is the least such encoding among its  $5,760$  images under  $G$ .

The supplement's `pair-orbits.json` has  $3,371$  entries in its `orbits` array, indexed by `orbit_index` from  $1$ . Each entry contains `R_masks`, `C_masks`, `stabilizer_order`, `orbit_size` and `stabilizer_maps`. A stabilizer map is the tuple  $(a, b, c, d, u, v, e)$  in (2); the array lists every element, including identity, rather than only generators. The stabilizer is

$$\text{Stab}_G(\{\mathcal{R}, \mathcal{C}\}) = \{g \in G : \{g\mathcal{R}, g\mathcal{C}\} = \{\mathcal{R}, \mathcal{C}\}\}.$$

It includes any maps exchanging the two partitions. The auxiliary field `pointwise_partition_stabilizer_order` counts maps fixing  $\mathcal{R}$  and  $\mathcal{C}$  individually, as partitions; it does not require fixing each block or point.

For example, the first orbit has

$$\begin{aligned}\mathcal{R} &= (53, 202, 13568, 51712), \\ \mathcal{C} &= (774, 3081, 20576, 41104).\end{aligned}$$

Its stabilizer consists of identity and  $(x, y) \mapsto (x + ty + 1, y)$ , and its orbit size is  $2,880$ . For illustration, the first mask  $53 = 1 + 4 + 16 + 32$  denotes the points  $(0, 0), (t, 0), (0, 1), (1, 1)$ . The remaining  $3,370$  representatives and their full stabilizers are supplied in the same format.

**Lemma 4.** *Let  $U_1, \dots, U_k$  be distinct valid canonical encodings of elements of  $\mathcal{S}$ , and let  $H_i$  be their full stabilizers. If*

$$\sum_{i=1}^k |G|/|H_i| = |\mathcal{S}|,$$

*then the list is a complete orbit classification.*

*Proof.* Two distinct canonical encodings cannot lie in one orbit, since an orbit has a unique least encoding. Thus the orbits of the  $U_i$  are disjoint subsets of  $\mathcal{S}$ . Orbit-stabilizer gives their cardinalities. Equality of the sum with  $|\mathcal{S}|$  forces their union to be  $\mathcal{S}$ .  $\square$

*Computer-assisted proof of Theorem 1.* For every listed pair, the verifier checks the block sizes, eligibility, disjointness, coverage, compatibility, increasing encodings, and absence of duplicate representatives. For every  $g \in G$ , it computes the canonical sorting of  $gU_i$  and checks that it is no smaller than  $U_i$ . It collects all  $g$  giving equality and compares this set of point permutations with the supplied stabilizer list. These checks cover all  $3,371 \cdot 5,760 = 19,416,960$  representative-map combinations.

All checks pass. They give exactly the stabilizer histogram in Table 1. Its orbit sizes sum to

$$\begin{aligned}3,229 \cdot 5,760 + 110 \cdot 2,880 + 23 \cdot 1,440 \\ + 7 \cdot 720 + 2 \cdot 360 = 18,954,720.\end{aligned}$$

This is the independently computed domain size in (4). Lemma 4 therefore proves completeness, in addition to validity, inequivalence and the full stabilizer assertions. The listed orbit counts sum to 3,371.  $\square$

There is no further factor of two in these orbit sizes: the objects are already unordered pairs. Equivalently, on ordered compatible pairs one can let  $G \times C_2$  act, with  $C_2$  exchanging the two coordinates. Each stabilizing point map of an unordered pair determines a unique choice of exchange in this ordered action. Stabilizer orders are therefore unchanged in that formulation, while orbit sizes double.

## 5 Construction and reproducibility

The certificate generator first reduces  $\mathcal{P}$  to its 17  $G$ -orbits. For each representative  $\mathcal{R}$  it enumerates compatible partners and their orbits under  $H = \text{Stab}_G(\mathcal{R})$ . These are precisely the ordered-pair  $G$ -orbits with first partition in  $G\mathcal{R}$ : an equivalence between  $(\mathcal{R}, \mathcal{C})$  and  $(\mathcal{R}, \mathcal{C}')$  must belong to  $H$ . This produces 6,662 ordered-pair orbits.

Exchange induces an involution on this set. To normalize an exchanged pair, transport its first partition to its partition-orbit representative and then normalize the second partition under the corresponding stabilizer. There are 80 fixed ordered-pair orbits, giving  $(6,662 + 80)/2 = 3,371$  unordered-pair orbits. This describes the construction of the certificate; the completeness proof uses the separate all-map verification and direct domain count.

The supplement contains the complete JSON list, its lossless text export, the Python generator and exporter, and a C++17 verifier. The text format begins with the number of representatives. Each entry then gives eight masks and a stabilizer order, followed by that many seven-parameter maps. The exported text was reconstructed from the JSON and compared byte for byte during verification.

After unpacking the supplement, a minimal independent replay is:

```
clang++ -std=c++17 -O3 \
    evidence/verify_classification.cpp \
    -o verify_classification
./verify_classification evidence/pair-certificate.txt
```

Exact commands for regenerating the certificate in a disposable copy are in the supplement's `README.md`. The generator uses Python 3.9 or later and its standard library. The executed C++ verifier was compiled with Apple clang 16.0.0 and uses integer masks and 64-bit counting totals. Neither program uses randomized sampling, floating-point mathematical tests, external solvers, or a theorem imported from the motivating paper.

A separate review implementation reconstructed field arithmetic, eligible blocks, partitions and semiaffine maps, read the JSON directly, and checked every representative and full stabilizer against all group elements. It also evaluated (5). The two verification routes agree on the complete census. This is an exact computer-assisted finite proof with inspectable code, not a proof-assistant formalization.

**Scope and assistance.** The result classifies the stated unlabeled partition pairs at order four. It makes no assertion for other orders or about isomorphism classes of labeled anti-Latin families. Historical priority and the detailed relationship to the motivating source remain unverified. OpenAI Codex assisted with mathematical exposition, algorithm design, code generation, execution and manuscript preparation. A separate Codex context performed the mathematical review and fresh implementation; this is a separate computational check, not external peer review or statistical independence of the reviewing systems.

## References

- [1] E. Aoyama, S. Hasegawa, M. Hayashi and T. Sagara, *Mutually orthogonal anti-Latin squares*, arXiv:2608.30082. Bibliographic record supplied with the motivating problem; primary text and metadata not independently verified here.
- [2] Zijian Zeng, *Proof-Carrying Reductions for 4-Regular Girth-Nine Graphs of Excess Four*, 2026. doi:10.2139/ssrn.7374858.